

Framingham Heart Study

The dataset (`framingham.csv`) contains cardiovascular medical records.

Dataset Schema Reference

- **Demographics:** `male` (0=Female, 1=Male), `age` , `education`
- **Behavioral:** `currentSmoker` , `cigsPerDay`
- **Medical History:** `BPMeds` , `prevalentStroke` , `prevalentHyp` , `diabetes`
- **Physical & Blood Metrics:** `totChol` , `sysBP` , `diaBP` , `BMI` , `heartRate` , `glucose`
- **Target Variable:** `TenYearCHD` (0=No 10-year Risk, 1=High 10-year Risk of Coronary Heart Disease)

Tasks

Task 1: Dataset Loading & Preliminary Health Check

Load `framingham.csv` into a Pandas DataFrame `df` . Perform an initial audit by printing:

1. The shape of the dataset.
2. The summary of non-null counts and data types for each column using `.info()` .
3. The count of missing values per column.

Task 2: Domain-Logic Missing Value Imputation

In medical datasets, missing values must follow clinical logic.

1. Inspect rows where `currentSmoker == 0` but `cigsPerDay` is `NaN` . Fill `cigsPerDay` with `0` for non-smokers.
2. For numeric continuous metrics (`totChol` , `BMI` , `heartRate` , `glucose`), fill missing values with the **median** of their respective columns.
3. For categorical columns (`BPMeds` , `education`), fill missing values with the **mode** (most frequent value).

Task 3: Value Mapping & Readability Transformation

The dataset uses `0` and `1` for the `male` column.

1. Map `male` to human-readable strings: `0 -> 'Female'` , `1 -> 'Male'` .

2. Rename the column `male` to `gender` .
3. Verify the class distribution of `gender` using `.value_counts()` .

Task 4: Extracting Risk Cohorts (`loc` and Slicing)

A medical researcher wants to isolate high-risk hypertensive patients.

1. Filter patients who have `sysBP >= 140` **OR** `diaBP >= 90` (clinical threshold for hypertension).
2. Select only the columns `['age', 'gender', 'totChol', 'sysBP', 'diaBP', 'TenYearCHD']` for these patients using `.loc[]` .
3. Save this subset into a new DataFrame `hypertensive_df` .

Task 5: Feature Engineering & Discretization (Binning)

Create a new categorical column `BMI_Category` based on standard WHO thresholds:

- Underweight : `BMI < 18.5`
- Normal : `18.5 < BMI < 25.0`
- Overweight : `25.0 < BMI < 30.0`
- Obese : `BMI > 30.0` Use `pd.cut()` to complete this discretization.

Task 6: Multi-Feature Grouped Statistics (`groupby`)

Analyze cardiovascular health metrics across gender and smoking status. Group the DataFrame by `gender` and `currentSmoker` and compute:

- Mean `totChol`
- Median `sysBP`
- Maximum `cigsPerDay`
- Mean `TenYearCHD` (risk percentage)

Task 7: Encoding Categorical Variables

Machine learning models require numerical inputs.

1. Apply One-Hot Encoding (`pd.get_dummies`) on the `BMI_Category` column created in Task 5.
2. Ensure the resulting dummy columns use binary numeric values (`0` and `1` or `True / False`).

Task 8: Outlier Detection & Trimming

Extremely high blood glucose levels can skew model training.

1. Calculate the 99th percentile value for the `glucose` column.
2. Identify and filter out records where `glucose` exceeds this 99th percentile threshold.

Task 9: Deriving Mean Arterial Pressure (Custom Feature Engineering)

Medical literature uses **Mean Arterial Pressure (MAP)** to estimate organ perfusion:

$$\text{MAP} = \text{diaBP} + \frac{1}{3} (\text{sysBP} - \text{diaBP})$$

Create a new feature column `MAP` using vectorized operations in Pandas.

Task 10: Model Matrix Preparation (Feature Matrix X and Target Vector y)

Prepare the final cleaned dataset for Scikit-Learn model input:

1. Separate target variable `TenYearCHD` into a 1D Pandas Series `y`.
2. Drop `TenYearCHD` from the DataFrame to form the feature matrix `x`.
3. Convert both `x` and `y` to NumPy arrays using `.values` and print their shapes.